

KAIKOURA, New Zealand

WMO: 936780

Lat: 42.4204S

Lon: 173.6950E

Elev: 354

StdP: 14.51

Time Zone: 12.00 (NZL)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.3	39.5	23.4	17.9	49.7	26.1	20.3	49.8	38.3	48.6	33.9	46.1	14.8	280	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.5	76.9	60.9	74.0	60.2	71.5	59.4	63.8	71.3	62.6	69.7	61.5	67.9	8.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	82.3	65.4	60.2	78.8	64.3	59.1	75.8	63.3	29.4	71.6	28.4	70.0	27.6	68.2	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
35.4	29.9	25.5	DB	35.7	84.4	1.5	2.5	34.6	86.2	33.8	87.7	32.9	89.1	31.8	91.0
			WB	32.8	66.5	1.5	1.4	31.7	67.5	30.9	68.4	30.1	69.2	29.0	70.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	61.0	61.1	59.2	55.2	52.3	48.8	47.1	48.0	50.4	52.6	55.1	59.0	(6)
	DBStd	6.83	5.20	4.71	4.89	4.48	4.81	4.32	4.23	4.42	4.94	5.07	5.19	5.32	(7)
	HDD50	408	0	0	0	7	28	74	112	94	54	28	10	1	(8)
	HDD65	4046	149	127	192	295	394	487	554	528	438	385	299	198	(9)
	CDD50	1913	341	311	287	162	99	38	23	31	66	109	164	282	(10)
	CDD65	73	24	18	13	1	0	0	0	0	0	1	2	14	(11)
	CDH74	246	89	62	32	1	1	0	0	0	0	3	10	48	(12)
	CDH80	18	7	4	3	0	0	0	0	0	0	0	0	4	(13)
(14) Wind	WSAvg	10.0	10.7	10.1	9.6	9.3	9.7	10.2	9.8	9.9	9.9	10.4	10.6	10.2	(14)
(15) Precipitation	PrecAvg	37.3	2.4	2.7	3.5	3.5	2.8	3.2	4.3	3.4	2.8	3.0	3.0	2.7	(15)
	PrecMax	54.3	9.6	5.9	7.5	11.0	8.7	12.8	16.8	13.5	7.8	5.6	8.4	9.3	(16)
	PrecMin	23.7	0.6	0.9	0.4	0.6	0.0	1.1	0.8	0.1	0.7	0.4	0.9	0.7	(17)
	PrecStd	7.2	1.8	1.3	2.1	2.2	1.7	2.3	3.5	2.9	1.9	1.3	1.6	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.6	80.0	77.4	71.9	70.3	64.4	62.6	64.1	68.0	71.8	74.5	79.2	(19)
		MCWB	62.0	62.9	61.1	57.1	55.1	50.8	49.5	50.4	52.5	55.2	57.2	60.9	(20)
	2%	DB	76.3	75.6	73.6	67.8	65.9	60.9	58.6	59.7	64.0	66.6	69.9	74.4	(21)
		MCWB	61.0	61.8	60.4	56.0	53.4	49.7	48.0	48.3	51.1	53.9	56.5	59.7	(22)
	5%	DB	73.2	72.5	70.5	64.9	62.6	58.2	56.4	57.0	61.2	63.8	67.0	70.9	(23)
		MCWB	60.2	61.2	59.0	54.6	52.2	48.4	46.8	47.2	50.0	52.6	55.9	59.1	(24)
	10%	DB	70.0	69.2	67.6	62.5	59.6	55.7	54.0	54.7	58.7	61.2	64.2	68.0	(25)
		MCWB	59.8	60.2	57.9	53.6	50.9	47.5	45.8	46.2	48.7	51.9	54.8	58.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	65.7	63.9	60.3	57.9	53.8	52.6	52.3	55.2	57.9	60.5	64.4	(27)
		MCDB	73.0	72.9	72.5	65.8	65.1	58.8	58.2	60.0	64.3	65.8	67.5	72.6	(28)
	2%	WB	63.7	63.9	62.0	58.3	55.7	51.9	50.2	50.3	53.0	55.8	58.8	62.2	(29)
		MCDB	70.8	71.1	69.4	63.7	60.6	56.4	54.6	55.3	60.4	63.2	65.8	69.7	(30)
	5%	WB	62.3	62.6	60.8	56.7	54.2	50.6	48.6	49.0	51.4	54.4	57.5	61.0	(31)
		MCDB	69.5	69.0	67.1	61.5	59.1	55.1	53.0	53.4	57.9	61.3	64.0	67.7	(32)
	10%	WB	61.1	61.5	59.5	55.4	52.7	49.2	47.3	47.9	50.2	53.0	56.1	59.7	(33)
		MCDB	67.3	66.8	64.9	59.7	57.5	53.6	51.8	52.0	55.9	59.0	62.2	65.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	10.5	10.1	9.2	8.4	8.2	7.7	8.1	9.7	10.8	11.0	10.9	(35)
		MCDBR	17.0	16.3	15.8	13.8	13.7	12.5	11.6	12.2	14.2	15.6	16.3	16.4	(36)
	5% WB	MCWBR	7.8	7.1	7.3	7.1	7.1	6.8	7.0	7.1	7.4	8.0	8.2	7.3	(37)
		MCWBR	13.6	13.3	12.5	11.0	10.2	9.3	9.0	9.4	12.1	13.4	13.9	13.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.323	0.309	0.297	0.286	0.280	0.275	0.288	0.301	0.315	0.316	0.323	(40)
		taud	2.596	2.640	2.675	2.687	2.671	2.651	2.678	2.665	2.602	2.587	2.613	2.631	(41)
	Ebn at Noon		316	310	301	284	265	255	267	283	298	309	319	320	(42)
		Edh at Noon	32	29	26	22	18	17	18	22	27	30	31	31	(43)
(44) All-Sky Solar Radiation	RadAvg		2056	1824	1423	944	639	481	559	809	1194	1632	2000	2050	(44)
	RadStd		164	171	75	79	39	29	38	55	68	101	154	145	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	N/A
(47) Regional (6 neighbors)		N/A	N/A	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page